DIOPHANTINE ANALYSIS ON HOMOGENEOUS VARIETIES AND CHARACTER VARIETIES (AND SOME APPLICATIONS), SPRING 2020 TAUGHT BY PETER SARNAK

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1. 2/4

The aim is to introduce basic tools to produce integer solutions, or to limit the number of solutions. What we mean by this is to find a solution $x = (x_1, \cdots, x_n) \in \mathbb{Z}^n$ which solves a system of equations $f_i(x_1, \cdots, x_n) \in \mathbb{Z}[x_1, \cdots, x_n]$. It is known that there is no procedure to decide the solutions of a general equation over $\mathbb{Z}$, by Matusevich (following Robinson–Davis).

There are obvious first obstructions in solving $f(x) = 0$ over $\mathbb{Z}$; this must be solvable over many other rings:

- $\mathbb{R}$;
- $\mathbb{Z}/n\mathbb{Z}$ for any integer $n \geq 1$;

- $\mathbb{Z}_p$ for any prime p.

If the equation passes all these "local tests", we can then hope that there is an integer solution. There are certain cases where this is automatic, and we will say that the equation satisfies the **local-to-global principle**, or the **Hasse principle**, if this holds.

These are a tentative list of specific topics we plan to cover.

(1) Circle method, and nilmanifolds replacing circle (for linear equations; by Furstenberg), both about counting solutions (given there are many solutions).
(2) Kloosterman method, levelling process.
(3) Solving quadratic forms: automorphic forms, spectral theory, trace formula (relating mod p solutions for different p's), Hasse–Minkowski, Siegel mass formula, subconvexity.
(4) Character varieties, using symmetry.
(5) Thue–Siegel–Roth, Schmidt.

Theorem 1.1 (Lang's $\mathbb{G}_m$)**.** *Let $V \subset (\mathbb{C}^\times)^N = T$ be an algebraic subvariety, and Γ be a finitely generated subgroup of T. Let $\overline{\Gamma}$ be the division group of Γ. Then, there are finitely many translates of algebraic subtori $T_1, \cdots, T_\ell$ (ℓ is effective) such that*

$$\overline{\Gamma} \cap V = \overline{\Gamma} \cap (T_1 \cup \cdots \cup T_\ell).$$

1.1. **Summation formulae.** Recall that the **Poisson summation formula** says that, for $f \in \mathcal{S}(\mathbb{R})$,

$$\sum_{m\in\mathbb{Z}} f(m) = \sum_{n\in\mathbb{Z}} \hat{f}(n).$$

The proof that everyone knows uses a periodic function

$$F(x) = \sum_{n\in\mathbb{Z}} f(x+n).$$

Then by Fourier expansion

$$F(0) = \sum_{m\in\mathbb{Z}} f(m) = \sum_{n\in\mathbb{Z}} \widehat{F}(n),$$

where

$$\widehat{F}(m) = \int_0^1 F(x)e(-xm)dx = \int_{-\infty}^{\infty} f(x)e(-mx)dx = \hat{f}(m) \text{ (basic case of Rankin–Selberg).}$$

Another way of writing this is to use a tempered distribution (i.e. bounded functional on the space of Schwartz functions, with Frechet topology ...)

$$\mu = \sum_{m\in\mathbb{Z}} \delta_m = \hat{\mu}.$$

Another way of proving is to use trace. Let $K(x,y) \in C^\infty(\mathbb{R}/\mathbb{Z} \times \mathbb{R}/\mathbb{Z})$ defined by $K(x,y) = \sum_{n\in\mathbb{Z}} f(x-y+n)$. Define

$$T_K\psi(x) = \int_0^1 K(x,y)\psi(y)dy.$$

Then K is a compact operator, so in particular is trace class. So the trace of it is on one hand

$$\mathrm{tr}(K) = \int_0^1 K(x,x)dx = \sum f(n),$$

and on the other hand it is the sum of eigenvalues, where eigenfunctions can be given by $e(-my)$;

$$\int K(x,y)e(-my)dy = \int \sum f(x-y+n)e(-my)dy = \hat{f}(m)e(-mx),$$

so $\sum f(n) = \sum \hat{f}(m)$. This also generalizes to automorphic forms, homogeneous spaces and discrete groups.

1.2. **Riemann–Weil–Guinand explicit formula.** This gives analytic continuation of Riemann zeta function without much hassle. From Poisson summaion, we get

$$\sum_n f(nx) = \frac{1}{x}\sum_m \hat{f}(\frac{m}{x}),$$

given $f \in \mathcal{S}(\mathbb{R})$, with $f(0) = \hat{f}(0) = 0$. Then

$$\int_0^\infty (\sum_n f(nx))x^s \frac{dx}{x},$$

is entire, and is the same as

$$2(\sum n^{-s})\int_0^\infty f(x)x^s\frac{dx}{x} = 2\zeta(s)M(f).$$

So by taking f appropriately one sees immediately that $\zeta(s)$ can be analytically continued, except for pole at $s = 1$. Also, if you put the other side of the Poisson summation, you get the functional equation!

From the Euler product, for $\mathrm{Re}(s) > 1$, we have

$$-\frac{\zeta'}{\zeta}(s) = \sum_{n=1}^\infty \lambda(n)n^{-s},$$

where $\lambda(n) = \begin{cases} \log p & \text{if } n = p^e \\ 0 & \text{otherwise} \end{cases}$. For an entire function $\Phi(s)$ rapidly decreasing in t ($s = \sigma + it$),

$$\frac{1}{2\pi i}\int_{\mathrm{Re}(s)=2}\Phi(s)\left(-\frac{\zeta'}{\zeta}\right)(s)ds = \sum_n \lambda(n)\frac{1}{2\pi i}\int_{\mathrm{Re}(s)=2}\Phi(s)n^{-s}ds.$$

Let $\frac{1}{2} + i\gamma$ be zeroes of $\Lambda(s) = \Gamma(\frac{s}{2})\pi^{-s/2}\zeta(s)$. Then for $g \in C_c^\infty(\mathbb{R})$, $h(r) = \int_{-\infty}^\infty g(u)e^{iru}du$,

$$\sum_\gamma h(\gamma) - h(i/2) - h(-i/2) = \frac{1}{2\pi}\int h(r)\left(\frac{\Gamma_\mathbb{R}'}{\Gamma_\mathbb{R}}(\frac{1}{2} + ir)\right) + \frac{\Gamma_\mathbb{R}'}{\Gamma_\mathbb{R}}(\frac{1}{2} - ir)dr - \sum_{n=1}^\infty \frac{\lambda(n)}{\sqrt{n}}(g(\log n) + g(-\log n)).$$

2. 2/6

Let $\Gamma \leq \mathrm{PSL}_2(\mathbb{R})$ be a cocompact discrete torsion-free group. Let $X = \Gamma\backslash\mathbb{H}$, which is a compact hyperbolic surface. Then

$$\mathrm{trace}(e^{i\sqrt{\Delta - \frac{1}{4}}t}),$$

is a tempered distribution in $t \in \mathbb{R}$, where $\Delta = \mathrm{div}\,\mathrm{grad}$.

Let ϕ_j be an eigenfunction of Δ, $\Delta\phi_j + \lambda_j\phi_j = 0$, which forms an orthonormal basis of functions on X. WLOG we can order λ_j's in an increasing order, so that $\lambda_0 = 0$, which corresponds to a constant function, and $\lambda_j = \frac{1}{4} + t_j^2$. Then there are finitely many eigenvalues less than or equal to $\frac{1}{4}$, and as $\lambda_j \longrightarrow \infty$, the eigenvalue grows like cj, for some constant c.

Conjecture 2.1 (Selberg's 1/4 conjecture)**.** *If Γ is arithmetic, then $\lambda_1 \geq \frac{1}{4}$.*

Remark 2.1. (1) Arithmeticity is needed, because as one goes up to cyclic cover you can make λ_1 arbitrarily small.

(2) We have $\frac{1}{4}$ because that is the smallest number in the spectrum of the universal cover, $\mathbb{H}$.

The set P of closed geodesics on X is in 1-1 correspondence with conjugacy classes of elements in Γ. This is because we have no elliptic or parabolic elements in Γ by assumption. Let $\ell(p)$ be the length of the closed geodesic corresponding to p, and $\ell_0(p)$ be the primitive length (primitive closed geodesic means it is not some closed geodesic rotated multiple times around). Namely the geodesic corresponding to $\begin{pmatrix} \lambda & 0 \\ 0 & \lambda^{-1} \end{pmatrix}$ is the part of vertical line connecting i and $\lambda^2 i$, so the length is $2\log\lambda$.

Theorem 2.1 (Selberg trace formula)**.** *Let g be even of compact support, and h be its Fourier transform. Then*

$$\sum_j h(t_j) = \frac{\mathrm{Area}(X)}{4\pi}\int_{-\infty}^{\infty} r\tanh(\pi r)h(r)dr + \sum_{p\in P}\frac{\ell(p_0)}{e^{\ell(p)/2}-e^{-\ell(p)/2}}g(\ell(p)).$$

2.1. **Crystalline measures.** Note that one can try to mimic this in a p-adic symmetric space, where for $\mathbb{H}^2$, the analogous space would be a Bruhat-Tits tree. A p-adic locally symmetric space would be a finite $(p+1)$-regular graph, and it can be shown that we get the zeta function of a variety over a finite field if we do the sam thing (Ihara).

We can also think of doing this on a **metric graph** Γ. This is just a compact 1-dimensional Riemannian manifold with finitely many singularities, with length assigned to each edge. The Laplacian would be $\frac{d^2}{dx^2}$ on the edges. One imposes the **Neumann boundary condition**, that ϕ should be continuous at the vertices, and that for all vertices v,

$$\sum_{e \text{ edge ending at } v} \partial_{\vec{e}_v}\phi(v) = 0.$$

Then what is $\mathrm{Spec}(\Gamma)$? It is an arithmetic progression if $\Gamma = S^1$, and it is an arithmetic progression for many other metric graphs, but there are also metric graphs whose spectra are very far from being arithmetic progression.

Definition 2.1. *A purely atomic measure μ on $\mathbb{R}$ is a* ***crystalline measure*** *if the support Λ is discrete, μ is a tempered distribution and $\hat{\mu}$ is also a purely atomic measure of discrete support.*

Theorem 2.2 (Meyer)**.** *Suppose $\mu = \sum_{\lambda\in\Lambda} a_\lambda\delta_\lambda$ is a crystalline measure such that $a_\lambda \in B$ for a finite set B. Suppose $|\hat{\mu}|$ is translation-bounded, $\sup_x |\hat{\mu}|(x+[0,1]) < \infty$, then μ is a "Dirac comb," i.e. an analogue of Poisson summation holds.*

Theorem 2.3 (Cohen idempotent theorem)**.** *For a measure μ on S^1 such that $\mu * \mu = \mu$, then the support of $\hat{\mu}$ is an arithmetic progression, up to a finite set.*

Theorem 2.4 (Littlewood's conjecture)**.** *For $k_1 < k_2 < \cdots < k_n$ integers,*

$$F_n(\alpha) = \sum_{j=1}^{\infty} e(\alpha k_j),$$

periodic on S^1, has L^1 norm growing at least like $O(\log n)$, with the lower bound growth rate achieved when k_i's are in arithmetic progression.

2.2. **Circle method.** Let $f(x_1, \cdots, x_s)$ be a nonsingular homogeneous polynomial of degree k with integral coefficients, and let $V_t := \{f(x) = t\}$. **Assume that $V_t(\mathbb{R})$ is noncompact** (so that there are many many solutions to expect). How rich is $V_t(\mathbb{Z})$?

Let N be large, and let

$$R_t(N) = \#\{f(x) = t \mid |x_j| \leq N\}.$$

As $N \longrightarrow \infty$, how does $R_t(N)$ behave? The points in the region, which we call B_N, is $(2N+1)^s$. Roughly, $|V_t(\mathbb{Z}) \cap B_N| \approx N^{s-k}$, but it can be just empty if there are congruence obstructions. That means the constant that is suppressed should contain local information ("local masses"). Also, it needs $s \gg k$ to be effective.

Theorem 2.5 (Heath-Brown, Browning). *If $s > 2^k$, then $R_t(N) \sim \mu_\infty(B_N) \prod_p \delta_t(p)$, where μ_∞ is the Hardy–Littlewood measure, supported on $V_t(\mathbb{R})$,*

$$\mu_\infty(K) = \lim_{\epsilon \longrightarrow 0} \frac{|\{x \in K \mid |f(x) - t| < \epsilon\}|}{2\epsilon},$$

and $\delta_t(p)$ is a local mass at p,

$$\delta_t(p) = \lim_{\nu \longrightarrow \infty} \frac{\# V_t(\mathbb{Z}/p^\nu \mathbb{Z})}{p^{\nu(s-1)}}.$$

3. 2/11

3.1. **Hardy–Littlewood circle method.** A general setting is as follows.

- $f(x_1, \cdots, x_s)$ is a homogeneous smooth polynomial of degree k.
- We want to count the integer solutions of $V_t = \{f(x) = t\}$.
- We have to have many variables, e.g. $s > 2^k + 1$, and we need V_t to be noncompact.
- There is no local obstruction, so that $V_t(\mathbb{Z})$ is nonempty and is expected to have many points.

  First assume f is diagonal,

  $$f(x) = b_1 x_1^k + \cdots + b_s x_s^k,$$

  with $b_1, \cdots, b_s \neq 0$.

We would then be counting

$$|V_t(\mathbb{Z}) \cap B_N|,$$

where B_N is a box of points whose coordinates are all of absolute value $\leq N$. We then consider a function like

$$F_N(\alpha) = \sum_{x \in \mathbb{Z}^s, |x_i| \leq N} e(\alpha F(x)),$$

for $\alpha \in \mathbb{R}/\mathbb{Z}$.

The reason why we consider this kind of exponential sum is because of Hardy–Ramanujan's work on estimates of partition function. Namely, to estimate the partition function, one can instead consider integral

$$\int_{0+iy}^{1+iy} \eta(z) e(-nz) dz,$$

and use the pole of η at cusps. This type of method is found succesful for generating series that is not even a modular form.

Indeed,

$$\int_0^1 F_N(\alpha)d\alpha = \#(B_N \cap V_0(\mathbb{Z})).$$

Then, $F_N(0) = (2N+1)^s$, which is the maximum value of $|F_N(s)|$, and for $s = a/q$, $(a,q) = 1$, $0 < a < q$,

$$F_N(a/q) = (2N/q)^s \sum_{x \bmod q} e(af(x)/q) + \text{smaller term} \sim (2N/q)^s t_{a,q}.$$

If we use that f is diagonal,

$$F_N(\alpha) = S_1(\alpha) \cdots S_s(\alpha),$$

where

$$S_j(\alpha) = \sum_{|x| \le N} e(b_j x^k \alpha).$$

Such one variable sum is called the **Weyl sum**. Weyl was interested in this because it appears in the analysis of $\zeta(1+it)$ after Taylor-expanding it.

Theorem 3.1 (Weyl's inequality)**.** *Let*

$$S(\alpha) = \sum_{|x| \le Q} e(\phi(x)),$$

where $\phi(x) = \alpha x^k + \alpha_1 x^{k-1} + \cdots + \alpha_k$ is a polynomial, α is an irrational number with $|\alpha - \frac{a}{q}| \le \frac{1}{q^2}$,

$$S(\alpha) \underset{\epsilon}{\ll} Q^{1+\epsilon}(q^{-1} + Q^{-1} + qQ^{-k})^{1/K},$$

where $K = 2^{k-1}$.

Proof. The strategy:

Square repeatedly reducing the degree of ϕ in inner sums and stop when ϕ is linear.

For example, consider estimating $S = \sum_{|x| \le Q} e(\alpha x^2)$. Then

$$|S|^2 = \sum_{x_1, x_2} e(\alpha(x_2^2 - x_1^2)) = \sum_{h < 2Q} (\text{some simple factor}) \sum_{x_1 \in I_h} e(\alpha((x_1+h)^2 - x_1^2)),$$

where we substitute using $x_2 = x_1 + h$. Thus we get a geometric series inside, and it can each be estimated by $\min(Q, \frac{1}{\|2h\alpha\|})$, where $\|x\|$ is the distance to the closest integer. Thus

$$|S|^2 \ll \sum_{h \le Q} \min(Q, \frac{1}{\|2h\alpha\|}).$$

Break the h-sum into progressions mod q, $Aq \le h \le (A+q)q$ with $h = Aq + r$. Letting $\alpha = \frac{a}{q} + \theta$, $|\theta| \le \frac{1}{q^2}$, $h\alpha = (Aq+r)(\frac{a}{q} + \theta)$, so after disregarding the case when $\frac{ra}{q} \equiv 0 (\bmod 1)$, the r-sum can be estimated by

$$\sum_{1 \le r' \le q/2} \frac{q}{r'} < q^{1+\epsilon}.$$

From this we get

$$|S|^2 \ll Q^{1+\epsilon} + Q^2/q,$$

which is the Weyl's inequality for $k = 2$. The full proof works in the same way with more bookkeeping. □

Exercise 3.1. Prove that if $\alpha \notin \mathbb{Q}$ then $\{x^k \alpha (\bmod 1)\}_{x \in \mathbb{N}}$ is equidistributed with respect to the Lebesgue measure on $[0,1]$.

Now we will define major and minor arcs.

Definition 3.1 (Major and minor arcs)**.** *Suppose q runs over $q \le N^\nu$ for small ν (say $\nu = \frac{1}{100}$). Define*

$$\mathscr{M}(q,a) = \left\{ \alpha \mid |\alpha - \frac{a}{q}| \le N^{\nu-k} \right\}.$$

Then,

$$\{\mathscr{M}(q,a)\}_{q \le N^\nu, \frac{a}{q} \text{ distinct rationals}},$$

are all disjoint intervals, and we call these arcs ***major arcs****. The complement m of $\mathscr{M} = \cup_{q \le N^\nu} \mathscr{M}(q,a)$ are called the* ***minor arcs****.*

We will see that when we split the integral

$$\int_0^1 F_N(\alpha) d\alpha = \int_{\mathscr{M}} + \int_m,$$

then we get the major term N^{s-k} from the integral over major arcs, and the rest would be small.

3.2. **Estimating minor arcs.** For $\alpha \in m$, for any $q \le N^\nu$,

$$\left| \alpha - \frac{a}{q} \right| \ge N^{\nu-k}.$$

On the other hand, by Dirichlet's box principle, there is $q^{k-\nu}$ such that

$$\left| \frac{a}{q} - \alpha \right| \le \frac{1}{qN^{k-\nu}}.$$

Thus, there is a, q such that $q > N^\nu$ and

$$\left| \frac{a}{q} - \alpha \right| \le \frac{1}{q^2}.$$

Let's for simplicity assume $s = 2^k + 1$. Then by Hua's lemma,

$$\int_0^1 |S(\alpha)|^{2^j} d\alpha \ll N^{2^j - j + \epsilon},$$

which is just an induction and Parseval's lemma. Then $F_N(\alpha) = S_1(\alpha) S_2(\alpha) \cdots S_s(\alpha)$, so

$$\int_m |F_N(\alpha)| d\alpha = \int_m |S_1(\alpha) \cdots S_s(\alpha)| d\alpha \le \max_{\alpha \in m} |S_1(\alpha)| \int_0^1 |S_1(\alpha)|^{2^k} d\alpha \ll N^{s-k-\delta},$$

which is a combination of Weyl's inequality and Hua's lemma.

4. 2/13

No class on next Thursday (2/20), office hours on Tuesdays at 3:30 in the tea room.

4.1. **Estimating major arcs (singular series and singular integral).** We are slightly changing our notation. Let's say

$$S_N(\alpha) = \sum_{|m|\le N} e(\alpha m^k).$$

Let's say $\alpha = \frac{a}{q} + \beta$, with $|\beta| \le N^{\nu-k}$. Then

$$\begin{aligned} S_N\left(\frac{a}{q}+\beta\right) &= \sum_{|m|\le N} e\left(\left(\frac{a}{q}+\beta\right)m^k\right) \\ &= \sum_{|m|\le N} e\left(\frac{am^k}{q}\right) e(\beta m^k) \\ &= \sum_{r=1}^{q} e\left(\frac{ar^k}{q}\right) \sum_{\lambda\le \frac{N}{q}} e(\beta(r+\lambda q)^k). \end{aligned}$$

Let's say $S(a,q) = \sum_{\lambda\le\frac{N}{q}} e(\beta(r+\lambda q)^k)$. Then

$$S_N(\alpha) = \frac{S(a,q)}{q}\int_{-N}^{N} e(\beta x^k)dx + O(N^{2\nu}).$$

Thus,

$$\int_{|\beta|\le N^{\nu-k}} F_N(\frac{a}{q}+\beta)d\beta \sim \frac{S_1(a,q)S_2(a,q)\cdots S_s(a,q)}{q^s}\int_{|\beta|\le N^{\nu-k}}\int_{-N}^{N}\cdots\int_{-N}^{N} e(\beta f(x))dxd\beta,$$

for a polynomial f. Letting $S(q,a) = \sum_{x(\mathrm{mod}\, q)} e\left(\frac{af(x)}{q}\right)$ and $S(q) = \sum_{(a,q)=1}\sum_{x(\mathrm{mod}\, q)} e\left(\frac{f(x)}{q}\right)$, this approximation becomes

$$\frac{S(q,a)N^{s-k}}{q^s}\int_{-N^\nu}^{N^\nu}\int_{-1}^{1}\cdots\int_{-1}^{1} e(\beta f(x))dx_1\cdots dx_s d\beta \sim N^{s-k}\sum_{q=1}^{\infty}\frac{S(q)}{q^s}\int_{-\infty}^{\infty}\int_{-1}^{1}\cdots\int_{-1}^{1} e(\beta f(x))dxd\beta,$$

after change of variables. The iterated integral is called **the singular integral**. We have proven the following

Theorem 4.1 (Hardy–Littlewood)**.** *We have*

$$\#\{V_0(\mathbb{Z})\cap B_N\} = N^{s-k}\sum_{q=1}^{\infty}\frac{S(q)}{q^s}\int_{-\infty}^{\infty}\int_{-1}^{1}\cdots\int_{-1}^{1} e(\beta f(x))dxd\beta + O(N^{s-k-\delta'}),$$

for $\delta' > 0$.

The series $\sum_{q=1}^{\infty}\frac{S(q)}{q^s}$ is called the **singular series**. This will have Euler product!

Exercise 4.1. Show that the archimedean local mass is

$$\mu_\infty(B_N) \sim N^{s-k}\int_{-\infty}^{\infty}\int_{-1}^{1}\cdots\int_{-1}^{1} e(\beta f(x))dxd\beta.$$

Show that the singular series is the product of all p-adic local masses,

$$\sum_{q=1}^{\infty}\frac{S(q)}{q^s} = \prod_p \delta_p(V_0).$$

Remark 4.1. This is a very basic result, and people in the area usually try to reduce the number of variables needed. For the case of diagonals the best is a method of Vinogradov (e.g. [Va]). More specifically, one replaces Hua's lemma with counting solutions of

$$x_1^j + \cdots + x_s^j = y_1^j + \cdots + y_s^j,$$

for $|x|, |y| \le N$, $1 \le j \le k$, using Vinogradov mean value theorem, decoupling, $\cdots$. The best so far is something like k^2, instead of $2^k + 1$.

If V is cut out by ν equations, then one could use the circle (rather torus in this case) method in the same way, but one needs many more variables to make it work.

One can also try to solve linear equations with extra restrictions that x_j's are primes. For example, twin prime conjecture or Goldbach's conjecture will fall into this category. Similar method could work, e.g. local to global principle, by Vinogradov, which can show that a large enough odd prime can be expressed as a sum of three primes.

Another way is Furstenberg's method using **nilmanifolds** in his proof of Szemeredi's theorem, followed by Green–Tao–Ziegler.

4.2. Applying the circle method.

Definition 4.1. *A variety cut out by $f_1, \cdots, f_\nu$ is **Hardy–Littlewood** if*

$$\#\{V(\mathbb{Z}) \cap B_N\} \sim \prod_{p \le \infty} (\textit{Hardy–Littlewood local densities}).$$

*It is **strongly Hardy–Littlewood** if the same holds with any congruence conditoins.*

*A variety V over $\mathbb{Z}$ is said to have **strong approximation property** if, for any $q > 1$,*

$$V(\mathbb{Z}) \twoheadrightarrow V(\mathbb{Z}/q\mathbb{Z}),$$

or that $V(\mathbb{Z}) \hookrightarrow V(\mathbb{Z}_p)$ is dense in p-adic topology for any prime p.

Remark 4.2. The definition is dependent on the set of equations using to define the variety. If the variety is complete intersection one can make sense of independent of defining equations.

Theorem 4.2. *If a variety is strongly Hardy–Littlewood, the variety has strong approximation property.*

An obvious question would be: which varieties are Hardy–Littlewood? There are a few ways of showing it, and in particular homogeneous varieties for linear actions can be understood using group theory and automorphic forms.

Exercise 4.2. Let $S(\alpha) = \sum_{x \le T} e(\alpha x^2)$. Show that

$$\int_0^1 |S(\alpha)| d\alpha \gg T^{\frac{1}{2} - \epsilon}.$$

Hint. Consider the L^4-norm and L^2-norm and compare them with the L^1-norm.

5. 2/18

5.1. Kloosterman method.

As there isn't more than square-root cancellation for most α for $S(\alpha) = \sum_{m \le M} e(\alpha m^k)$, one cannot alllow minor arcs in for example $k = 2, s = 4$ case. In this case we can instead use the **Kloosterman method**, which uses Farey points of level Q, namely $\frac{a}{q}$ with $q \le Q$, $(a, q) = 1$, $1 \le a \le q - 1$. One then splits $\int_0^1$ into small intervals with endpoints

Farey points. A smooth version is done by Duke–Friedlander–Iwaniec. One should also look for Heath-Brown, Hooley.

What Kloosterman method could do is to show that a diagonal quadratic form in four variables,

$$f(x_1, \cdots, x_4) = a_1 x_1^2 + \cdots + a_4 x_4^2,$$

is strongly Hardy–Littlewood. In the process, the famous **Kloosterman sum** (which is secretly a finite field analogue of Bessel function) appears,

$$S(m, n; c) = \sum_{x \bmod c, x\overline{x} \equiv 1 (\bmod c)} e\left(\frac{mx + n\overline{x}}{c}\right),$$

and one needs to estimate the sum. Obviously oe has $|S(m, n; c)| \leq c$ but one needs more.

By Chinese remainder theorem, if $(c_1, c_2) = 1$,

$$S(m, n; c_1 c_2) = S(\overline{c}_2 m, \overline{c}_2 n, c_1) S(\overline{c}_1 m, \overline{c}_1 n, c_2),$$

so this is not multiplicative; thus, the series

$$Z(s) = \sum_c \frac{S(m, n; c)}{c^s},$$

is not an Euler product. Nevertheless it will have analytic continuation and will be related to modular forms in a very deep way.

To estimate, it is enough to estimate $S(m, n; p^\ell)$. By linearizing, one can reduce to the case $\ell = 1$. Kloosterman after all obtained an estimate

$$S(m, n; p) \ll p^{3/4}.$$

Obviously one can exclude $(m, n) \neq (0, 0) (\bmod p)$.

Exercise 5.1. Show

$$\sum_{m \bmod p} |S(m, l; p)|^4 = p^3 + \text{smaller},$$

and show the Kloosterman's bound.

Hasse realized, and Weil proved, the Weil conjectures for Artin–Schreier curves, which yields

$$|S(m, n; p)| \leq 2\sqrt{p}.$$

Exercise 5.2. Using Weil's bound, show that

$$|S(m, n; c)| \ll_\epsilon c^{1/2+\epsilon},$$

for $(m, n, c) = 1$.

Remark 5.1. For homogeneous equations, one can push through the Kloosterman method + circle method to get down to three variables, and prove Hasse–Minkowski theorem (over $\mathbb{Q}$).

Unfortunately the local-to-global principle does not hold for $f(x_1, x_2, x_3) = t$ for $t \neq 0$ and f an integral quadratic form. We will treat this using automorphic forms.

5.2. **Siegel mass formula.** Let F be a quadratic form, integral over over a number field K. We would like to deal with $F(x) = t$, which could be thought as $X^t AX = t$. One could also consider a non-scalar version, $X^t AX = B$, for symmetric integral matrices A and B. Now the key point is that $G = \mathrm{O}(F)$ acts linearly on the solutions! We can consider $G(\mathbb{Z})$, which acts on the set of integral solutions, and $G(\mathbb{Z}) \leq G(\mathbb{R})$ is big discrete subgroup such that $\mathrm{vol}(G(\mathbb{Z})\backslash G(\mathbb{R})) < \infty$, if $n \geq 3$. If $n = 2$, there can be exceptions, which includes Pell's equations.

Definition 5.1. *Following Gauss, integral quadratic forms F_1, F_2 are in the same genus if they are locally integrally equivalent. Integral quadratic forms F_1, F_2 are integrally equivalent if $F_1(x) = F_2(\gamma x)$ for some $\gamma \in \mathrm{GL}_n(\mathbb{Z})$.*

Theorem 5.1 (Gauss, Hermite, Minkowski, Smith)**.** *A genus of integral quadratic forms decomposes into a finite number h (the **class number of the genus**) integrally inequivalent forms.*

The main reason behind this is that there is a transitive action by a reductive linear algebraic group. More generally, if $V = G/H$, then except for obvious reasons $V(\mathbb{Z})$ has finitely many $G(\mathbb{Z})$-orbits. This uses geometry of numbers (Minkowski's reduction theory). For example:

Theorem 5.2 (Mahler compactness)**.** *A subset $\mathscr{L} \subset \mathrm{GL}_n(\mathbb{Z})\backslash \mathrm{GL}_n(\mathbb{R})$ is precompact if and only if the length of the shortest nonzero vector in any lattice $L \in \mathscr{L}$ is bounded below by a positive constant. All volumes are bounded above by the constant.*

One nice reference is [Ca].

6. 2/25

6.1. **Siegel mass formula.** We recall the setting. Let $F(x_1, \cdots, x_n)$ be an integral quadratic form, and suppose $n \geq 4$; otherwise there are factors of 2 that come from stabilizers that are not osemisimple. Suppose that F is definite at ∞. Define $r_F(m)$ be the number of representations of $m > 0$ such that $F(x) = m$.

Let $F \sim F_1, \cdots, F_h$ be inequivalent classes in the genus (h is the class number of the genus). Then it turns out that $r_{F_j}(m)$ considered together is much easier:

Theorem 6.1 (Siegel mass formula)**.**

$$\frac{\sum_{j=1}^{h} \frac{r_{F_j}(m)}{|\mathrm{O}_{F_j}(\mathbb{Z})|}}{\sum_{j=1}^{h} \frac{1}{|\mathrm{O}_{F_j}(\mathbb{Z})|}} = \mu_\infty(V_m(\mathbb{R})) \prod_{p<\infty} \delta_p(V_m).$$

Remark 6.1. If $h = 1$, then this gives a local-to-global principle over $\mathbb{Z}$. But it is extremely rare that $h = 1$.

Example 6.1. Consider $F(x_1, x_2) = x_1^2 + x_2^2$ and consider $V_t(\mathbb{Z})$ when t is prime. In this case $h = 1$, and one can compute the local masses,

$$\delta_2(V_t) = \begin{cases} 0 & \text{if } t \equiv 3 (\mathrm{mod}\, 4) \\ 2 & \text{if } t \equiv 1 (\mathrm{mod}\, 4) \end{cases}$$

and for $p \neq 2$,

$$\delta_p(V_t) = \begin{cases} \left(1 - \left(\frac{-1}{p}\right) p^{-1}\right) & \text{if } p \neq t \\ 2\left(1 - \left(\frac{-1}{t}\right) t^{-1}\right) & \text{if } p = t \end{cases}$$

and

$$\mu_\infty = \pi.$$

So, for $t \equiv 1 (\bmod 4)$,

$$r_F(t) = 2\pi \prod_{p>2} \left(1 - \left(\frac{-1}{p}\right) p^{-1}\right) = 8.$$

The quadratic forms $F(x_1, x_2, x_3) = x_1^2 + x_2^2 + x_3^2$ and $F(x_1, x_2, x_3, x_4) = x_1^2 + x_2^2 + x_3^2 + x_4^2$ both have class number 1.

6.2. **Dirichlet class number formula.** What about an indefinite quadratic form? Consider an indefinite form

$$F(x_1, x_2, x_3) = x_2^2 - 4x_1x_3.$$

Then $V_d(\mathbb{Z})$ decomposes into a finite number of orbits of $\mathrm{O}_F(\mathbb{Z})$. However in this case $\mathrm{O}_F(\mathbb{Z})$ is an infinite group; $\mathrm{O}_F(\mathbb{R})/\mathrm{O}_F(\mathbb{Z})$ is finite volume.

Theorem 6.2 (Dirichlet class number formula)**.** *Let $d > 0$ be squarefree, and consider $b^2 - 4ac = d$. Then*

$$h(d) \log \epsilon_d = (*)|d|^{1/2} L(1, \chi_d),$$

where $(*)$ is some archimedean volume factor (like π or sth), $h(d)$ is the number of orbits, and ϵ_d is the solution of $t^2 - du^2 = 4$.

Write

$$V_t(\mathbb{Z}) = \coprod_{j=1}^{\nu} x_j \, \mathrm{O}_F(\mathbb{Z}).$$

Let H_{x_j} be the stabilizer of x_j in O_F. Note that $V_t(\mathbb{R}) \cong G(\mathbb{R})/H_{x_j}(\mathbb{R})$ where $G = \mathrm{O}_F$. The mass formula for an indefinite form for $n \geq 3$ involves $\mathrm{Vol}(H_{x_j}(\mathbb{Z})\backslash H_{x_j}(\mathbb{R}))$ and $\mathrm{Vol}(G(\mathbb{Z})\backslash G(\mathbb{R}))$. We choose the Haar measures on G and H_{x_j} so that it is consistent with the Hardy–Littlewood measure μ_∞ on V_t.

Theorem 6.3 (Siegel's formula)**.** *Suppose that F is indefinite, the discriminant of F is squarefree (so $n \geq 3$ then $h = 1$). Let ν be the number of $\mathrm{O}_F(\mathbb{Z})$-orbits of $V_t(\mathbb{Z})$. Then*

$$\sum_{j=1}^{\nu} \frac{\mathrm{Vol}(H_{x_j}(\mathbb{Z})\backslash H_{x_j}(\mathbb{R}))}{\mathrm{Vol}(G(\mathbb{Z})\backslash G(\mathbb{R}))} = \prod_p \delta_p(V_t).$$

When $n = 3$ and F is $x_2^2 - 4x_1x_3$, this is exactly the Dirichlet's formula.

Recall that the Hardy–Littlewood method says, for $n \geq 4$,

$$|B_T \cap V_t(\mathbb{Z})| \sim \mu_\infty(B_T) \prod_p \delta_p(V_t).$$

How are they related? This is the content of Eskin–Rudnick–Sarnak.

Remember how we counted points inside a big ball B; it is roughly $\mathrm{Vol}(B)$. For $B \subset \mathrm{O}_F(\mathbb{R})$ a "big round blob", with lattice $\Gamma \leq \mathrm{O}_F(\mathbb{R}) =: G(\mathbb{R})$, we want to say that

$$|B \cap \Gamma| \sim \frac{\mathrm{Vol}(B)}{\mathrm{Vol}(\Gamma\backslash G)}.$$

But this is not so easy as in the Euclidean space. In the Euclidean case, we could easily aprpoximate with the volume because the boundary area is of lower order than the volume. But in the hyperbolic case this is not true; for example, in the upper half plane, a ball of radius R is of area e^R but its perimeter is also of length e^R!

7.1. **Duke–Rudnick–Sarnak.**

Theorem 7.1 (Duke–Rudnick–Sarnak)**.** *Let F be an integral quadratic form. Let $G = \mathrm{O}_F$, and let V_t be the solution space of $F(x) = t$. Fix $x_0 \in V_t$, and let H_{x_0} be the stabilizer of x_0 in G. Then as $T \longrightarrow \infty$,*

$$\{x_0\gamma \mid \gamma \in \mathrm{O}_F(\mathbb{Z}), |x_0\gamma| \leq T\} \sim \mu_\infty(B_T)\frac{\mathrm{Vol}(H_{x_0}(\mathbb{Z})\backslash H_{x_0}(\mathbb{R}))}{\mathrm{Vol}(G(\mathbb{Z})\backslash G(\mathbb{R}))},$$

where μ_∞ is the Hardy–Littlewood measure and the Haar measures are compatible with respect to μ_∞.

From this we can deduce the Siegel's formula Theorem 6.3, which is a special case of the Siegel mass formula.

Proof of Siegel's formula. We sketch several proofs.

(1) Proof of Siegel, e.g. in 1943 Tata lectures (?? I couldn't find a book). This elementary proof uses the Hasse–Minkowski theorem.
(2) Proof of Siegel, using Siegel modular forms, theta functions, etc. This does not use the Hasse principle but rather deduces the Hasse principle along the way. This is where he came up with the Siegel–Weil theorem, which says in modern language that "adelic integral of a theta function against 1 is an Eisenstein series."
(3) Proof of Weil, using Tamagawa numbers. Tamagawa's insight was that all of Siegel's formulae are equivalent to $\tau(\mathrm{SO}_F) = 2$, for $n \geq 3$. This statement is proven without referring to prior proofs in [We].

    Tamagawa's observation was as follows. Let $G = \mathrm{SO}_F$, $n \geq 3$. Then $G(\mathbb{Q})\backslash G(\mathbb{A})$ is of finite volume, and whenever you choose Haar measure at each place using an algebraic volume form of G over $\mathbb{Q}$, then the volume is actually independent of scaling, because the effect of scaling by a rational number cancels out over all places. This volume is called the **Tamagawa number** $\tau(G)$. Tamagawa showed $\tau(G) = 2$ is equivalent to all Siegel formulae. To be more descriptive, a formal calculation gives something with a factor of $\tau(\mathrm{SO}_F)/\tau(\mathrm{SO}_H)$, where H is the stabilizer. Then you can start doing an induction; the base case is that F is four variable, H is three variable, both indefinite. Then we know $\tau(\mathrm{SO}_H) = 2$ which is Dirichlet's result, and we know the Siegel's formula by different method, so $\tau(\mathrm{SO}_F) = 2$. Going over and over, we now know that $\tau(\mathrm{SO}_F) = 2$ for all indefinite F of 3 or more variables. From this we can get that the Tamagawa number for definite orthogonal groups is 2 because one can find definite orthogonal stabilizer inside indefinite orthogonal group.
(4) Similarly, one can deduce that the Tamagawa number is 2 if you know the Weil's conjecture on Tamagawa numbers that a simply connected $\mathbb{Q}$-group has Tamagawa number 1. This is proved by Kottwitz using trace formula. This uses Hasse principle.

□

7.2. **Modular forms.** The most interesting modular form in our setting would be the theta function

$$\theta(z) = \sum_{m=-\infty}^{\infty} e(m^2 z),$$

for $z \in \mathfrak{h}$.

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